

Closed-form blackboard response and estimation from randomized game logs

Companion technical note

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Abstract

The propensity-weighted response estimator and the solvable blackboard model describe the same intervention-versus-silence contrast under the model assumptions. We derive their connection, give a simple approximation and describe estimation using existing round and microscopic logs. No new population experiment or full transition-density estimator is required for the initial comparison.

1 What the randomized score measures

Let H_t contain all pre-action history, including the sampled votes Y_t . Let U_t be the sampled binary controller activation and $e_t = \Pr(U_t = 1 \mid H_t)$ its recorded probability. With a fixed designated target, write $D_t = x_{t+1} - x_t$ and

$$R_t = \left[\frac{U_t}{e_t} - \frac{1 - U_t}{1 - e_t} \right] D_t. \quad (1)$$

Assume correct sequential randomization given H_t , $0 < e_t < 1$, consistent outcome timing, and that each episode's outcome is unaffected by other episodes. Agent interactions within an episode are allowed. The gate must use the recorded probability; if it depends only on Y_t , the logged $\Pr(U_t = 1 \mid Y_t)$ is sufficient. Then

$$\mathbb{E}[R_t \mid H_t] = \mathbb{E}[D_t \mid U_t = 1, H_t] - \mathbb{E}[D_t \mid U_t = 0, H_t] \quad (2)$$

$$= \mathbb{E}[x_{t+1}(1) - x_{t+1}(0) \mid H_t] \equiv \tau_t(H_t). \quad (3)$$

The factors e_t cancel on taking the expectation; the shared pre-action x_t cancels too. Thus the average score estimates the immediate effect of activating the controller over the histories encountered by the experiment. It is not the effect of always intervening throughout an episode. Memory does not invalidate this identity, and x_t need not be Markovian. The policy still affects which histories are encountered and the estimator's variance.

For any pre-action summary S_t ,

$$\mathbb{E}[R_t \mid S_t] = \mathbb{E}[\tau_t(H_t) \mid S_t]. \quad (4)$$

Keep failed deliveries and fallbacks in their sampled $U_t = 1$ arm. This targets activation of the implemented communication policy, including its failures. Do not condition the primary causal contrast on realized message mode, exposure, acquisitions or actual post count: these occur after activation.

2 Closed form under the simple blackboard model

Hold the task, evidence regime and response parameters fixed during a round. Let B be a common fixed eligible peer-message count, b the controller posts under activation, and q the sample size ($q \leq B$ here). Assume binary exposure is sufficient and independent of the current vote. Then

$$w_b = 1 - \frac{\binom{B}{q}}{\binom{B+b}{q}}. \quad (5)$$

If fewer than q total messages exist, replace q by $\min(q, B + b)$ with the conventions in the preceding model note. Budget equals b only if fully used.

At a microscopic update, an unexposed agent revises with probability γ_0 to a Bernoulli(p_0) target vote; an exposed agent uses (γ_1, p_1) . Both branches include baseline reasoning. The exposure mixture has

$$\gamma_b = (1 - w_b)\gamma_0 + w_b\gamma_1, \quad p_b = \frac{(1 - w_b)\gamma_0 p_0 + w_b\gamma_1 p_1}{\gamma_b}. \quad (6)$$

For M uniform agent updates with replacement, set

$$L_u = 1 - (1 - \gamma_u/N)^M, \quad u \in \{0, b\}. \quad (7)$$

The model's mean displacement is $(p_u - x)L_u$. Combining this with Eq. (3) gives the requested closed form:

$$\boxed{\mathbb{E}[R_t \mid x_t = x, b] = \chi_b(x) = L_b(p_b - x) - L_0(p_0 - x) = A_b + C_b x,} \quad (8)$$

where $A_b = L_b p_b - L_0 p_0$ and $C_b = L_0 - L_b$. For $\gamma_u = 0$, the corresponding term is zero. There is no extra multiplication by e_t : Eq. (8) is the activation-versus-silence effect.

Simplest approximation. If $\gamma_0 = \gamma_1 = \gamma$, then

$$\chi_b = w_b(p_1 - p_0)[1 - (1 - \gamma/N)^M] \simeq \left[1 - \left(\frac{B}{B+b}\right)^q\right] (p_1 - p_0)(1 - e^{-\gamma M/N}). \quad (9)$$

The first approximation requires sampling a small fraction of the board; the exponential approximation requires small γ/N with $M\gamma^2/N^2$ small. For $M = N$, the last factor is approximately $1 - e^{-\gamma}$. The finite- N expression is inexpensive and should be preferred in computations.

Averaging matters. With fixed parameters, $\mathbb{E}[R \mid b] = A_b + C_b \mathbb{E}[x \mid b]$. With heterogeneous histories, average the history-specific predicted responses instead; plugging averaged parameters into a nonlinear expression is generally different. The constant-parameter result is a reference model, not a consequence of randomization.

3 Estimate a response curve before fitting mechanisms

Within a fixed task, target direction, persistence and budget, fit the score directly:

$$\hat{C}_b = \frac{\sum_i (x_i - \bar{x})(R_i - \bar{R})}{\sum_i (x_i - \bar{x})^2}, \quad \hat{A}_b = \bar{R} - \hat{C}_b \bar{x}. \quad (10)$$

This is a closed-form linear projection of the causal response; it estimates the model's coefficients if Eq. (8) is correct. It does not identify all four microscopic parameters. If x has insufficient variation, report the mean response only. Use held-out episodes or initialization blocks to assess prediction; do not use in-sample agreement as model validation.

4 How to use the current game records

The following fields were checked by decoding the existing report-only batch’s `rounds.parquet`. Its 6,720 controller-enabled round records contain propensities from 0.1192029220 to 0.8807970780. This confirms availability in that batch, not the schema or completeness of a newer run.

Quantity	Existing round field
U_t	<code>controller_sampled_U</code>
e_t	<code>P_U1_given_Y</code>
x_t	<code>controller_target_share_before</code>
x_{t+1}	<code>controller_target_share</code>
Assigned allowance	<code>intervention_budget</code>
Episode and time	<code>episode_key, round_index</code>

Primary calculation. Verify the two shares use the same controller target and bracket the intended round. Use both $U = 0$ and $U = 1$ rounds from controller-enabled runs with valid propensities; standalone no-control episodes are not propensity-weighted observations. Compute Eq. (1) once per round. Report its mean by prespecified strata, and display means in the existing eight x bins with counts for both actions and uncertainty. Keep task, target direction, persistence and round index separate initially. Any aggregation must state its weights; equal-task or equal-episode weighting is not automatic row pooling.

Resample independent episodes, or whole shared-initialization/randomness blocks when runs are coupled. Do not treat rounds or agent updates as independent replicates. Preserve the same aggregation weights in every bootstrap replicate. Missing outcomes need an explicit audit; silently retaining only successful outcomes can bias the contrast. Individual scores can exceed $[-1, 1]$ even though the true response is bounded; do not clip them.

Optional mechanistic calibration. In `micro_slots.parquet`, use `focal_opinion_before/after` relative to `round_controller_target`. Define exposure $E = 1$ from nonempty `sampled_controller_message_ids`, rather than a legacy direct-control flag. For each exposure class j , estimate

$$a_j = \Pr(0 \rightarrow 1 \mid \text{initial vote } 0, E = j), \quad c_j = \Pr(1 \rightarrow 0 \mid \text{initial vote } 1, E = j). \quad (11)$$

Under the homogeneous reset model,

$$\hat{\gamma}_j = \hat{a}_j + \hat{c}_j, \quad \hat{p}_j = \frac{\hat{a}_j}{\hat{a}_j + \hat{c}_j}. \quad (12)$$

The denominators count eligible focal updates in each initial-vote class. Zero total switching leaves p_j unidentified; $\hat{a}_j + \hat{c}_j > 1$ violates this reset model or reflects sampling error. Check uncertainty rather than silently clipping. Exposure-based fits are descriptive unless their additional homogeneity assumptions hold; randomizing activation does not randomize realized exposure.

Board calibration and recommendation. Empirical exposure frequency and eligible-message counts diagnose w_b . A growing dawn-seeded board is not frozen: using its initially empty peer pool ($B = 0$) for the entire round is inappropriate. A fitted effective B is an approximation, and realized exposure is a mediator, not a causal stratum. Start with the score means and the two-coefficient fit; add microscopic calibration only to explain a discrepancy. Keep this causal response separate from the old observational T_π and η_{IR} unless all components are recomputed under a consistent joint distribution.

Context. This additive score identity follows directly from sequential randomization. Related causal-excursion methodology: Qian et al., *Estimating time-varying causal excursion effects in mobile health with binary outcomes*, [arXiv:1906.00528](#). Policy-dependent interpretation: Guo, Richardson and Robins (2021), [arXiv:2103.02323](#). The blackboard formulas above are derived from the proposed two-kernel reference model.